

Conversion of Units within the Metric System

Jonathan R. Bacolod

The **International System of Units (SI)** is the standard system of measurement used worldwide.

Larger Unit to Smaller Unit

To convert from a **larger unit** to a **smaller unit**, multiply by the appropriate power of 10 (move the decimal point to the **right**).

$$2 \text{ km} = 2 \times 1000 = 2000 \text{ m}$$

Smaller Unit to Larger Unit

To convert from a **smaller unit** to a **larger unit**, divide by the appropriate power of 10 (move the decimal point to the **left**).

$$5000 \text{ m} \div 1000 = 5 \text{ km}$$

Common Prefixes

Prefix	Value
kilo (k)	1000
hecto (h)	100
deka (da)	10
base unit	m, g, L
deci (d)	$\frac{1}{10}$
centi (c)	$\frac{1}{100}$
milli (m)	$\frac{1}{1000}$

The Conversion Ladder

A useful conversion ladder is:

k → **h** → **da** → **Unit** → **d** → **c** → **m**

Each step is equivalent to multiplying or dividing by 10.

Mnemonic: *King Henry Died Unusually
Drinking Chocolate Milk*

Examples

Example 1

Convert 3.6 km to m.

km $\rightarrow$ m is 3 steps down the ladder, so multiply by 1000.

$$3.6 \times 1000 = 3600 \text{ m}$$

Example 2

Convert 4500 g to kg.

g $\rightarrow$ kg is 3 steps up the ladder, so divide by 1000.

$$4500 \div 1000 = 4.5 \text{ kg}$$

Example 3

Convert 6.2 L to mL.

L $\rightarrow$ mL is 3 steps down the ladder, so multiply by 1000.

$$6.2 \times 1000 = 6200 \text{ mL}$$

Example 4

Convert 2500 cm to m.

cm $\rightarrow$ m is 2 steps up the ladder, so divide by 100.

$$2500 \div 100 = 25 \text{ m}$$

Practice Exercises

Practice

Convert each quantity to the indicated unit.

Practice 1

4.8 km to m

Practice 2

3600 cm to m

Practice 3

5.2 kg to g

Practice 4

8450 mL to L

Practice 5

7.25 m to cm

Practice 6

9800 mg to g

Practice 7

3.4 L to mL

Practice 8

5600 g to kg

Practice 9

0.85 m to mm

Practice 10

125000 mm to m.

Activities

Activity

Convert each quantity to the indicated unit.

Activity 1

6.3 km to m

Activity 2

4250 cm to m

Activity 3

8.4 kg to g

Activity 4

9600 mL to L

Activity 5

9.15 m to cm

Activity 6

12600 mg to g

Activity 7

2.75 L to mL

Activity 8

8700 g to kg

Activity 9

1.2 m to mm

Activity 10

98000 mm to m.

Thank you!